HSI–LiDAR Joint Classification via Progressive Spatial–Spectral–Frequency Fusion Learning

Xinxin Liu, *Member, IEEE*, Xiaoqing Tang, Ting Lu, *Member, IEEE*, and Kexin Ding

***Abstract*—Hyperspectral image (HSI) and light detection and ranging (LiDAR) data joint classification leverages multimodal information to improve classification performance. However, existing methods often suffer from inadequate feature representation due to their reliance on spatial–spectral domains while ignoring complementary frequency-domain cues, limiting their ability to capture cross-modal variations. Moreover, even when frequency information is incorporated, most approaches fail to achieve deep cross-domain interaction, either treating spatial and frequency features in isolation or relying on single-stage fusion, which hinders effective modality gap bridging. To address these challenges, we propose a novel progressive spatial–spectral–frequency fusion learning (PS2F2L) network, a lightweight yet powerful framework for HSI and LiDAR classification. Our method employs a multistage progressive fusion strategy to hierarchically integrate multimodal features, mitigating modal conflicts while capturing discriminative features. In the first stage, dual branches—spatial–spectral feature learning (S2FL) and spatial–frequency feature learning (SF2L)—jointly extract features from complementary domains, overcoming single-domain limitations. The S2FL branch combines 1-D/2-D convolutions to model spectral–spatial relationships, while SF2L utilizes discrete wavelet transforms (DWTs) to capture spatial–frequency patterns. In the second stage, an interactive spatial–spectral–frequency fusion module enhances feature discriminability by promoting deep information exchange between spatial–spectral and spatial–frequency representations. Finally, adaptive decision-level fusion (DLF) refines classification by consolidating multidomain predictions. Extensive experiments on three public datasets demonstrate the superiority of PS2F2L, validating its effectiveness in achieving robust and accurate multimodal classification.**

***Index Terms*—Hyperspectral images (HSIs), light detection and ranging (LiDAR) data, spatial–frequency feature fusion, supervised classification.**

I. Introduction

With the continuous development of remote sensing imaging technology, multisource data provide multidimensional observations for refined land cover monitoring [1]. For example, hyperspectral image (HSI) [2], [3], [4], [5] contains rich spectral information, which contributes to a deeper understanding of land material and composition. Light detection and ranging (LiDAR) data [6], [7] have anti-interference characteristics and can provide high-precision elevation information. These two data sources depict land cover features from distinct observational perspectives, exhibiting strong complementarity in feature representation. Therefore, combining HSI and LiDAR data can effectively overcome the limitations of single data sources and further improve the accuracy and reliability of land classification. This enables critical applications in multiple fields, including military reconnaissance [8], environmental monitoring [9], and urban planning [10]. Therefore, the joint classification of HSI and LiDAR has become a hotspot in contemporary remote sensing research.

In recent years, deep learning (DL) has gained significant attention for the joint classification of HSI and LiDAR data due to its excellent feature learning capabilities [11], [12], [13]. As a powerful feature extraction tool, convolutional neural networks (CNNs) have been widely adopted in classification tasks and have shown remarkable potential through various enhanced architectures [14], [15], [16], [17]. For example, Hang et al. [14] employed two coupled CNNs combined with a parameter sharing mechanism to achieve efficient fusion classification of HSI–LiDAR data. Multiscale convolutions [17] and dilated convolutions [18] were introduced to capture hierarchical spatial patterns of ground objects. Morphological convolution blocks applied by Roy et al. [15] and Li et al. [19] aimed to extract discriminative spatial features in HSI and LiDAR joint classification. The 3-D convolutions [20] have proven effective in performing joint convolutions across spatial and spectral dimensions. Despite advances in CNN-based feature extraction, the fusion of HSI and LiDAR information remains challenging due to their inherent modal differences. Conventional approaches like simple concatenation or elementwise weighted fusion often yield suboptimal classification performance. To address this problem, recent studies have developed more sophisticated fusion strategies, including attention mechanisms [21] and gated fusion [22], which better account for cross-modal complementary relationships. However, ground objects exhibit distinct representations across modalities, which cannot be adequately characterized using conventional image-domain feature extraction methods limited to spatial and/or spectral processing.

As a unique perspective for information analysis, frequency-domain analysis enables signal decomposition based on frequency components to extract and enhance classification-discriminative features [23]. Current research

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has explored frequency-domain approaches, which can be primarily categorized into Fourier-based and wavelet-based methods. Among them, the Fourier transform (FT) analyzes an image's entire frequency spectrum, which provides global pattern observation in the frequency domain. The FT directly computes global frequency representations without requiring progressive receptive field expansion. This capability provides effective support for modeling long-range dependence [24], [25], [26]. However, the FT's global nature lacks spatial frequency localization, hindering its application to the multimodal fusion task that requires local feature alignment. In contrast, the wavelet transform (WT) employs wavelet basis functions at varying scales, enabling localized analysis in both spatial and frequency domains. This allows WT to effectively capture critical structural features in remote sensing data, such as edges and textures, which are essential for the HSI–LiDAR joint classification. For example, Ni et al. [27] exploited 2-D and 3-D WTs to extract multiscale features, fully leveraging their advantages in multiscale analysis and localized feature representation. Yang et al. [28] used graph convolution and graph wavelet convolution operations to capture multiscale low-frequency and high-frequency features from HSI and LiDAR data. Although these studies attempt to improve multimodal image classification through frequency-domain analysis, most approaches only treat it as a supplementary tool for feature extraction or decision-making. A significant lack of joint exploration and coupled utilization of cross-domain information remains.

To overcome the limitations of single-domain representations and achieve substantial improvements in classification performance, multidomain information fusion [29], [30], [31] has emerged as a promising research direction. Among them, Shi et al. [29] proposed a multilevel feature gated fusion network with dual attention mechanisms in both spatial and frequency domains, to better capture global subtle features. Song et al. [31] designed a multiscale cross-domain fusion network that achieves enhanced classification by combining cross-modal information across different domains. The above studies fuse spatial- and frequency-domain features via cross-domain attention or multiscale integration, demonstrating the effectiveness of multidomain joint strategies in improving classification accuracy. Nevertheless, the existing single-stage fusion paradigms fundamentally limit cross-modal integration, failing to establish hierarchical information interaction or bridge the modality gap completely.

Although various DL methods for joint classification of HSI and LiDAR have been developed, there are still challenges that demand further research. First, most methods focus on spatial-domain feature extraction and fusion [15], [18], while neglecting complementary cues from frequency or other transform domains. This often leads to inadequate feature representation, as significant cross-modal feature variations cannot be effectively captured through single-domain processing. Second, some methods [25], [28] attempt to introduce frequency-domain information to supplement spatial-domain information, but these approaches maintain essentially isolated feature learning processes across different domains. Consequently, they only achieve unidirectional guidance or shallow integration of information, failing to fully exploit cross-domain correlations through joint feature exploration. Finally, although existing multidomain joint perception strategies [30], [31] have begun to explore interactions between spatial and frequency domains, most remain limited to single-step fusion approaches. These either fail to achieve deep cross-domain integration for bridging modality gaps or lack efficient network designs to accomplish such integration.

To address the above challenges, we designed a progressive spatial–spectral–frequency fusion learning (PS2F2L) network for joint classification of HSI and LiDAR data. Unlike single-step fusion, the proposed multistage progressive fusion architecture provides two key benefits. On the one hand, it maintains modality-specific features while mitigating intermodal conflicts to produce more stable and discriminative fused representations. On the other hand, it enables lightweight model design that achieves better classification performance with reduced parameters. More specifically, the first stage jointly extracts multimodal features from both spatial–spectral and spatial–frequency domains to overcome the limitations of single-domain perception. The second stage enhances cross-domain complementary features through interactive fusion of spatial–spectral and spatial–frequency representations, strategically utilizing interdomain discrepancies to boost multimodal feature learning. The final stage refines classification stability and reliability through decision-level fusion (DLF) of the enhanced multidomain features.

The main contributions of this work are summarized as follows.

1) A multistage progressive fusion framework is proposed for multimodal joint classification, which sequentially performs joint extraction of spatial–spectral/spatial–frequency multimodal features, cross-domain enhanced spatial–spectral–frequency fusion, and decision-level integration to provide more comprehensive fusion with optimal performance-efficiency tradeoffs.
2) A dual-branch feature learning architecture is designed to effectively integrate multimodal data, where 1-D/2-D convolutions jointly model spectral–spatial features while wavelet decomposition captures spatial–frequency characteristics, thereby obtaining discriminative joint representations from heterogeneous HSI and LiDAR data.
3) An interactive spatial–spectral–frequency fusion module is proposed to achieve deep integration across different domains. This module leverages feature differences as attention cues to selectively amplify complementary information and suppress redundant information, thus enhancing the reliability and stability of the classification process.

The remainder of this work is organized as follows. Section II introduces the related work. Section III details the proposed framework and its components. Section IV provides the experimental datasets, comparative analysis of different methods, and performance evaluation. The conclusion is given in Section V.

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II. Related Work

This section will focus on two topics related to this article: progressive learning and spatial–spectral–frequency fusion.

A. Progressive Learning

In the joint classification task of HSI and LiDAR data, modal fusion is a key technology for enhancing classification accuracy and robustness. This technology primarily encompasses three types: pixel-level fusion, feature-level fusion, and DLF, among which feature-level fusion is currently the most widely adopted approach. Early studies [14], [17], [21], [22] commonly employed CNNs for HSI–LiDAR classification. Methods that rely on straightforward concatenation or elementwise addition for HSI–LiDAR fusion largely overlook the inherent semantic heterogeneity across modalities, resulting in limited classification performance. To further enhance information exchange, more sophisticated fusion strategies have been developed, including attention mechanisms [21], [32] and gated fusion [22]. These methods adaptively weight or select cross-modal features to more effectively model the complementary relationship between HSI and LiDAR data. Nevertheless, most of them remain confined to a single fusion stage, lacking the capability for progressive modeling, which is crucial for comprehensively capturing multilevel semantic information, enhancing robustness against interference, and ensuring overall stability.

Some studies [1], [20], [33] have attempted to incorporate the transformer architecture for multimodal fusion. Although its self-attention mechanism can capture long-range dependencies, it often fails to fully account for modality-specific priors in HSI–LiDAR classification. Moreover, most designs remain confined to single cross-attention interactions, falling short of achieving true multistage collaboration.

Progressive learning is an effective strategy for handling multimodal tasks, with its core principle being the gradual increase in complexity or information volume throughout the learning process [34]. By guiding cross-modal information exchange in stages, it progressively bridges the representational gaps between multimodal images, thereby more effectively addressing the fusion challenges posed by modal heterogeneity. To this end, this article introduces a novel network termed PS2F2L for joint HSI–LiDAR classification, aiming to achieve superior accuracy through a carefully designed yet lightweight multistage fusion framework.

B. Spatial–Spectral–Frequency Fusion

Existing HSI–LiDAR joint classification methods primarily rely on spatial–spectral feature modeling [14], [15], [17], [18]. However, ground objects exhibit fundamental representational differences between HSI and LiDAR data. Traditional image-domain methods, constrained to processing spatial and/or spectral dimensions, struggle to comprehensively capture this cross-modal heterogeneity. As a crucial complementary perspective in information analysis, frequency-domain analysis effectively extracts and enhances latent features that are discriminative for classification by decomposing signals into different frequency components. Compared to the FT, which provides only global frequency information, the WT simultaneously analyzes frequency content and local spatial details, offering a more informative multiresolution representation. It can precisely capture key structural details in remote sensing data (such as edges, textures, and scale variations), which is crucial for distinguishing complex land cover categories [27], [31]. To capture essential information for classification from heterogeneous data, we combine a spatial–frequency representation [obtained via discrete WT (DWT)] with spatial–spectral features through a progressive fusion learning strategy. This enables the model's discriminative capability to be effectively enhanced.

III. Proposed Method

A. Overall Framework

To extract discriminative features from highly heterogeneous multimodal data, we develop a novel multistage progressive fusion framework that achieves deeper cross-modal integration than single-step approaches through lightweight yet effective network design. The proposed PS2F2L network comprises three successive fusion stages: initial multimodal feature extraction, followed by cross-domain enhancement, culminating in DLF. PS2F2L's overall architecture is shown in Fig. 1.

Specifically, in the first stage, we construct a dual-branch architecture including spatial–spectral feature learning (S2FL) and spatial–frequency feature learning (SF2L) to address single-domain learning constraints by jointly extracting features from HSI–LiDAR data. In the second stage, we design an interactive spatial–spectral–frequency fusion module to enhance multidomain complementary features by leveraging cross-domain difference information. In the third stage, the enhanced multidomain features undergo efficient DLF to obtain fine-grained classification. The proposed framework systematically enhances cross-modal classification performance by embedding well-designed fusion strategies throughout the "feature extraction–interactive enhancement–classification decision" pipeline.

Let $\mathbf{H} \in \mathbb{R}^{\mathbf{m}\times\mathbf{n}\times\mathbf{D}}$ and $\mathbf{L} \in \mathbb{R}^{\mathbf{m}\times\mathbf{n}\times\mathbf{d}}$ be the geographically registered hyperspectral and LiDAR images with a spatial size of $m \times n$, where $\mathbf{D}$ and $\mathbf{d}$ denote the number of channels in HSI and LiDAR images, respectively. To reduce spectral redundancy in HSI, principal component analysis (PCA) is used to project $\mathbf{H}$ onto its first $\mathbf{k}$ principal components, yielding a reduced representation as $\mathbf{H}' \in \mathbb{R}^{\mathbf{m}\times\mathbf{n}\times\mathbf{k}}$. For the convenience of network training, we then perform patch extraction by splitting $\mathbf{H}'$ and $\mathbf{L}$ into $\mathbf{p} \times \mathbf{p}$ small patches, denoted as $\mathbf{X^h} \in \mathbb{R}^{\mathbf{p}\times\mathbf{p}\times\mathbf{k}}$ and $\mathbf{X^l} \in \mathbb{R}^{\mathbf{p}\times\mathbf{p}\times\mathbf{d}}$, respectively.

B. Domain-Adaptive Feature Extraction via HSI–LiDAR Cross-Modal Fusion

In order to achieve effective coupling of multimodal information from different perceptual domains, we construct S2FL based on 1-D/2-D convolutions and SF2L based on DWTs. This design prevents excessive parameter growth during the initial fusion stage.

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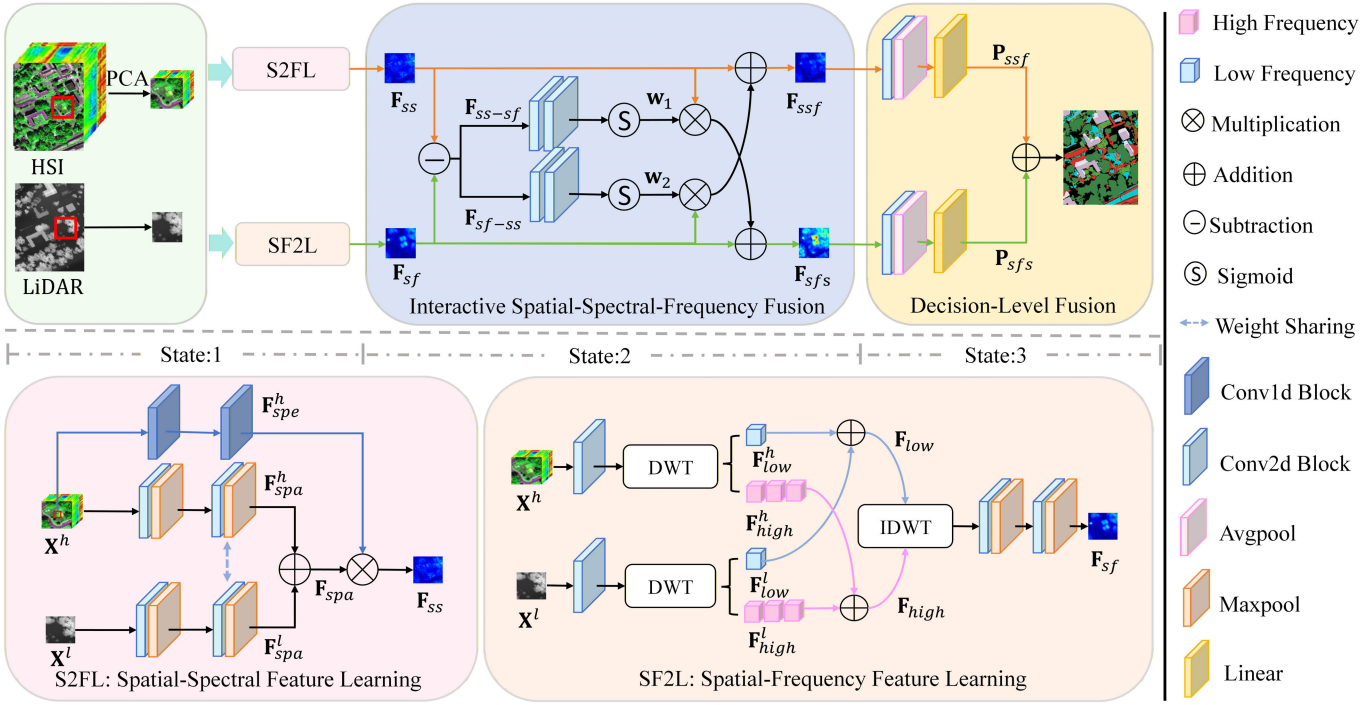


Fig. 1. Architecture of the proposed PS2F2L network for HSI–LiDAR joint classification.

1) Spatial–Spectral Feature Learning: Numerous studies have demonstrated [16], [35], [36] that 2-D convolutions excel at extracting image spatial features, while 1-D convolutions are better at capturing spectral features. Inspired by these methods, we use 2-D and 1-D convolutions to build an S2FL module. S2FL employs a three-branch structure, i.e., one spectral branch and two spatial branches. The spectral branch contains two sequential 1-D convolution blocks, each containing a 1-D convolution layer followed by batch normalization (BN) and rectified linear unit (ReLU) activation. The blocks utilize 32 and 64 convolution kernels, respectively, to enable more diverse feature extraction.

For computational efficiency, our spatial feature extraction branch employs only two 2-D convolution blocks with pooling layers for HSI and LiDAR processing. At shallow layers, convolutional weights adapt independently to capture modality-specific features, while deeper layers share weights to leverage their inherent semantic consistency for object recognition. As demonstrated in [14], this strategy not only significantly reduces the number of model parameters but also leverages inherent semantic consistency to enable coupled learning. This aligns closely with our objectives of achieving multimodal collaboration through lightweight model design. Specifically, each 2-D convolution block consists of a 3×3 convolution, BN, and ReLU activation function, with the number of convolution kernels matching the spectral branch. Subsequent 2×2 max pooling then downsamples features to maintain discriminative patterns with reduced dimensionality.

Ultimately, the deep features of both modalities are fused via addition to form multimodal spatial features. Compared to feature concatenation, feature addition can effectively fuse well-aligned semantic information without increasing the channel dimension, reduce redundant information interference, and simultaneously lower the computational burden on subsequent network layers, thereby improving the overall training and inference efficiency of the model. Let $\mathbf{F^h_{spa}}$, $\mathbf{F^l_{spa}}$, and $\mathbf{F_{spa}}$ denote the HSI, LiDAR, and fused spatial features, respectively. Then, the spatial feature fusion can be mathematically expressed as follows:

$$\mathbf{F_{spa}} = \mathbf{F^h_{spa}} + \mathbf{F^l_{spa}}. \tag{1}$$

Next, spatial–spectral feature fusion is achieved through feature multiplication. As an elementwise modulation mechanism, multiplication enables nonlinear interaction between spatial and spectral features. This allows spatial features to adaptively enhance or suppress relevant spectral features, enabling more selective feature fusion. The corresponding operation is formulated by

$$\mathbf{F_{ss}} = \mathbf{F^h_{spe}} \otimes \mathbf{F_{spa}} \tag{2}$$

where $\mathbf{F^h_{spe}}$ denotes the spectral features of HSI, $\mathbf{F_{ss}}$ denotes the fused spatial–spectral features, and $\otimes$ is the operator that performs elementwise multiplication of two tensors.

2) Spatial–Frequency Feature Learning: While spatial–spectral feature extraction effectively captures local structures and spectral characteristics of ground objects, it often struggles to represent complex textures and fine boundary details. To address this limitation, we develop SF2L with frequency-domain representations to better encode high-frequency components and structural patterns. This complementary strategy helps to enhance the model's discriminative capacity for complex ground objects.

Compared to the fast FT (FFT), which only provides global frequency information, the DWT can capture both frequency

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components and local spatial details. With its stronger positional awareness, DWT is more suitable for joint classification tasks. The core of DWT is the wavelet basis function, which consists of a scale function and a wavelet function. The db2 wavelet basis functions offer strong edge feature sensitivity and excellent computational efficiency. To this end, we apply the db2 wavelet basis in DWT to realize effective frequency-domain analysis of the feature signals.

The HSI and LiDAR data are first input into a 2-D convolution block, which contains a 2-D convolution (with 32 kernels), BN, and ReLU activation function. Next, the HSI- and LiDAR-related raw features are decomposed into one low-frequency and three high-frequency components by DWT. The low-frequency component reflects most of the background information, while the three high-frequency components indicate details in horizontal, vertical, and diagonal directions, respectively. The high-frequency and low-frequency components of the HSI and LiDAR data can be obtained through the equations as follows:

$$\mathbf{F}^{\mathbf{h}}_{\mathbf{low}}, \mathbf{F}^{\mathbf{h}}_{\mathbf{high}} = \mathbf{DWT}\left(\mathbf{Conv2d}\left(\mathbf{X}^{\mathbf{h}}\right)\right) \tag{3}$$

$$\mathbf{F}^{\mathbf{l}}_{\mathbf{low}}, \mathbf{F}^{\mathbf{l}}_{\mathbf{high}} = \mathbf{DWT}\left(\mathbf{Conv2d}\left(\mathbf{X}^{\mathbf{l}}\right)\right) \tag{4}$$

where $\mathbf{DWT}(\cdot)$ denotes the DWT operation. $\mathbf{F}^{\mathbf{h}}_{\mathbf{low}}$ and $\mathbf{F}^{\mathbf{h}}_{\mathbf{high}}$ represent the low- and high-frequency components of HSI data, respectively, while $\mathbf{F}^{\mathbf{l}}_{\mathbf{low}}$ and $\mathbf{F}^{\mathbf{l}}_{\mathbf{high}}$ correspond to the low- and high-frequency components of LiDAR data, respectively.

In order to exploit the complementary spatial–frequency information from HSI and LiDAR data, the high- and low-frequency components of both modalities are further fused by an addition operation, generating enhanced high- and low-frequency features according to the following equations:

$$\mathbf{F}_{\mathbf{low}} = \mathbf{F}^{\mathbf{h}}_{\mathbf{low}} + \mathbf{F}^{\mathbf{l}}_{\mathbf{low}} \tag{5}$$

$$\mathbf{F}_{\mathbf{high}} = \mathbf{F}^{\mathbf{h}}_{\mathbf{high}} + \mathbf{F}^{\mathbf{l}}_{\mathbf{high}} \tag{6}$$

where $\mathbf{F}_{\mathbf{low}}$ and $\mathbf{F}_{\mathbf{high}}$ denote the fused low- and high-frequency components, respectively. This fusion strategy helps to minimize redundant feature disturbance while keeping feature map dimensions unchanged.

The fused high- and low-frequency components are transformed back to the feature domain using the inverse DWT (IDWT). After that, it is processed through two identical 2-D convolution blocks to achieve cross-branch feature alignment (with spatial–spectral features) while enhancing the network's representational capacity. The number of convolution kernels for each 2-D convolution is set to 64. The corresponding computational process can be described

$$\mathbf{F}_{\mathbf{sf}} = \mathbf{Conv2d}\left(\mathbf{Conv2d}\left(\mathbf{IDWT}\left(\mathbf{F}_{\mathbf{low}}, \mathbf{F}_{\mathbf{high}}\right)\right)\right) \tag{7}$$

where $\mathbf{IDWT}(\cdot)$ denotes the IDWT operation, and $\mathbf{F}_{\mathbf{sf}}$ represents the output spatial–frequency features generated by module SF2L.

C. Cross-Domain Complementary Feature Enhancement via Interactive Feature Fusion

Domain-specific feature learning methods can effectively capture distinctive characteristics that may not be observable in other domains. The inherent complementarity between the features learned by our S2FL (spatial–spectral) and SF2L (spatial–frequency) modules motivates us to further perform feature fusion.

To promote effective complementary feature enhancement and cross-domain interactive fusion, we first calculate the difference characteristics of the spatial–spectral and spatial–frequency features. These features are then refined through an attention mechanism using two sequential 2-D convolution blocks and a sigmoid activation. The first 2-D convolution block contains a 2-D convolution with a kernel size of 3×3 and the LeakyReLU activation function. This activation function maintains a linear mapping with a slope of 1 for positive inputs, while assigning a smaller nonzero slope to negative inputs. This design ensures a nonzero gradient flow during backpropagation, effectively mitigating the vanishing gradient problem. The second 2-D convolution block contains only one 2-D convolution with a kernel size of 3×3. The number of kernels for these two 2-D convolutions is set to 32 and 64, respectively. The attention mechanism selectively filters information by dynamically weighting the importance of interdomain differential features. To realize cross-domain mutual enhancement, the attention weights are derived from bidirectional domain differences as follows:

$$\mathbf{w_1} = \mathbf{attn}\left(\mathbf{F_{ss}} - \mathbf{F_{sf}}\right) \tag{8}$$

$$\mathbf{w_2} = \mathbf{attn}\left(\mathbf{F_{sf}} - \mathbf{F_{ss}}\right) \tag{9}$$

where $\mathbf{attn}(\cdot)$ denotes the function of attention mechanism. $\mathbf{w_1}$ and $\mathbf{w_2}$ measure the importance of difference features for spatial–frequency and spatial–spectral enhancement, respectively.

These attention weights enable elementwise feature fusion to dynamically emphasize significant cross-domain characteristics. Through a compensation mechanism, we then reinforce the original $\mathbf{F_{sf}}$ and $\mathbf{F_{ss}}$ features by reinjecting selected discriminative information, formulated as

$$\mathbf{F_{sfs}} = \mathbf{w_1} \otimes \mathbf{F_{ss}} + \mathbf{F_{sf}} \tag{10}$$

$$\mathbf{F_{ssf}} = \mathbf{w_2} \otimes \mathbf{F_{sf}} + \mathbf{F_{ss}} \tag{11}$$

where $\mathbf{F_{ssf}}$ and $\mathbf{F_{sfs}}$ represent the mutually enhanced cross-domain features after spatial–spectral–frequency fusion.

D. Classification-Stability Refinement via DLF

While $\mathbf{F_{ssf}}$ and $\mathbf{F_{sfs}}$ features provide enhanced cross-domain representations, their individual classification performance remains limited due to inherent decision biases. To ensure robust classification, we introduce a DLF stage that performs independent class predictions from both features, followed by adaptive probabilistic fusion.

First, $\mathbf{F_{ssf}}$ and $\mathbf{F_{sfs}}$ are classified by a CNN-based classifier that contains a 2-D convolution block and a fully connected layer. The convolution block comprises two sequential layers. The first layer implements 1×1 convolution with 32 kernels, followed by BN and ReLU activation. The second layer performs adaptive global average pooling, reducing spatial dimensions to 1×1 to generate a compact 32-D patch-level feature vector. This vector aggregates the global

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TABLE I
HSI- AND LIDAR-BASED DSM DATASETS USED FOR EVALUATION

Dataset Name	Houston2013		Trento		MUUFL	
Location	Houston, Texas, USA		Trento, Italy		Long Beach, Mississippi, USA	
Data Type	HSI	LiDAR-based DSM	HSI	LiDAR-based DSM	HSI	LiDAR-based DSM
Sensor Name	CASI-1500	-	AISA Eagle	Optech ALTM 3100EA	CASI-1500	Gemini ALTM LiDAR
Spatial Resolution	2.5 m	2.5 m	1 m	1 m	0.54 m × 1.0 m	0.60 m × 0.78 m
Number of bands	144	1	63	1	64	2
Wavelength	0.38 - 1.05 μm	-	0.42 - 0.99 μm	-	0.38 - 1.05 μm	1.06 μm
Image Size	349×1905	349×1905	166×600	166×600	325 × 220	325 × 220
Number of categories	15		6		11	
Number of labeled samples	15029		30214		53687	

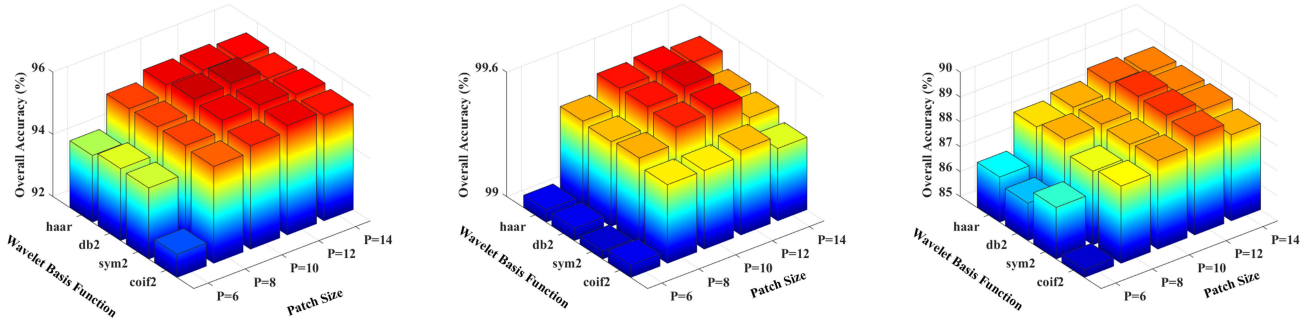


Fig. 2. Classification performance under different wavelet bases and patch sizes across datasets. (a) Houston2013. (b) Trento. (c) MUUFL.

contextual information of the input patch, representing the ground object category to which the central pixel belongs. Subsequently, the fully connected layer then transforms the feature channels into **C** dimensions, representing the target ground object categories. The classification probabilities $\mathbf{P_{ssf}}$ and $\mathbf{P_{sfs}}$ from the spatial–spectral–frequency features $\mathbf{F_{ssf}}$ and spatial–frequency–spectral features $\mathbf{F_{sfs}}$ are adaptively fused to boost accuracy. The final classification probability **P** can be calculated by

$$\mathbf{P} = \beta \mathbf{P_{ssf}} + (\mathbf{1} - \beta)\, \mathbf{P_{sfs}} \tag{12}$$

where $\boldsymbol{\beta}$ is an adaptive weighting factor between the two branches. It is initialized to 0, implying equal initial contributions. During training, $\boldsymbol{\beta}$ is adjusted in response to the classification accuracy of the $\mathbf{P_{ssf}}$ and $\mathbf{P_{sfs}}$ branches, thereby optimizing the ultimate classification accuracy.

E. Overall Loss Function

In order to obtain more reliable classification performance, we design a dual-constrained loss function that accommodates the subtle differences between two spatial–spectral–frequency feature learning processes. Each learning process is constrained by a cross-entropy loss function, which can be expressed as follows:

$$\mathcal{L}_{\mathbf{cls}} = \frac{\mathbf{1}}{\mathbf{M}} \sum_{\mathbf{i=1}}^{\mathbf{M}} \mathbf{y^{(i)}} \log\left(\mathbf{P^{(i)}}\right) + \left(\mathbf{1} - \mathbf{y^{(i)}}\right) \log\left(\mathbf{1} - \mathbf{P^{(i)}}\right) \tag{13}$$

where **M** is the number of samples in each batch, $\mathbf{P^{(i)}}$ is the predicted class probability of the **i**th sample, and $\mathbf{y^{(i)}}$ denotes its corresponding class label.

The final overall loss function $\mathcal{L}$ combining both the spatial–spectral–frequency categorical loss $\mathcal{L}_{\mathbf{cls}}^{\mathbf{ssf}}$ and the spatial–frequency–spectral categorical loss $\mathcal{L}_{\mathbf{cls}}^{\mathbf{sfs}}$ can be formulated

$$\mathcal{L} = \mathcal{L}_{\mathbf{cls}}^{\mathbf{ssf}} + \mathcal{L}_{\mathbf{cls}}^{\mathbf{sfs}}. \tag{14}$$

Through this dual-constrained overall loss function, the proposed model enables progressive spatial–spectral–frequency feature fusion for HSI–LiDAR data classification within an end-to-end learning paradigm.

IV. EXPERIMENTS AND ANALYSIS

In this section, we conduct comprehensive experiments on three widely used HSI and LiDAR benchmark datasets to demonstrate our method's advantages over the state-of-the-art (SOTA) approaches.

A. Experimental Setting

1) Dataset Illustration: Three multimodal datasets were used in the experiments, including Houston2013, Trento, and MUUFL datasets. These datasets primarily comprise HSI- and LiDAR-based digital surface models (DSMs). More detailed information is shown in Table I. Among them, the Houston2013 dataset captures a complex urban scene, containing diverse man-made ground objects and extensive vegetation coverage. It covers 15 land cover categories with a size of 349 × 1905 pixels. The Trento dataset represents a mixed urban and agricultural scene, with relatively uniform vegetation distribution. It has a size of 166 × 600 pixels and covers six land cover categories. The 325 × 220 pixels MUUFL dataset represents a campus scene, which includes 11 land cover

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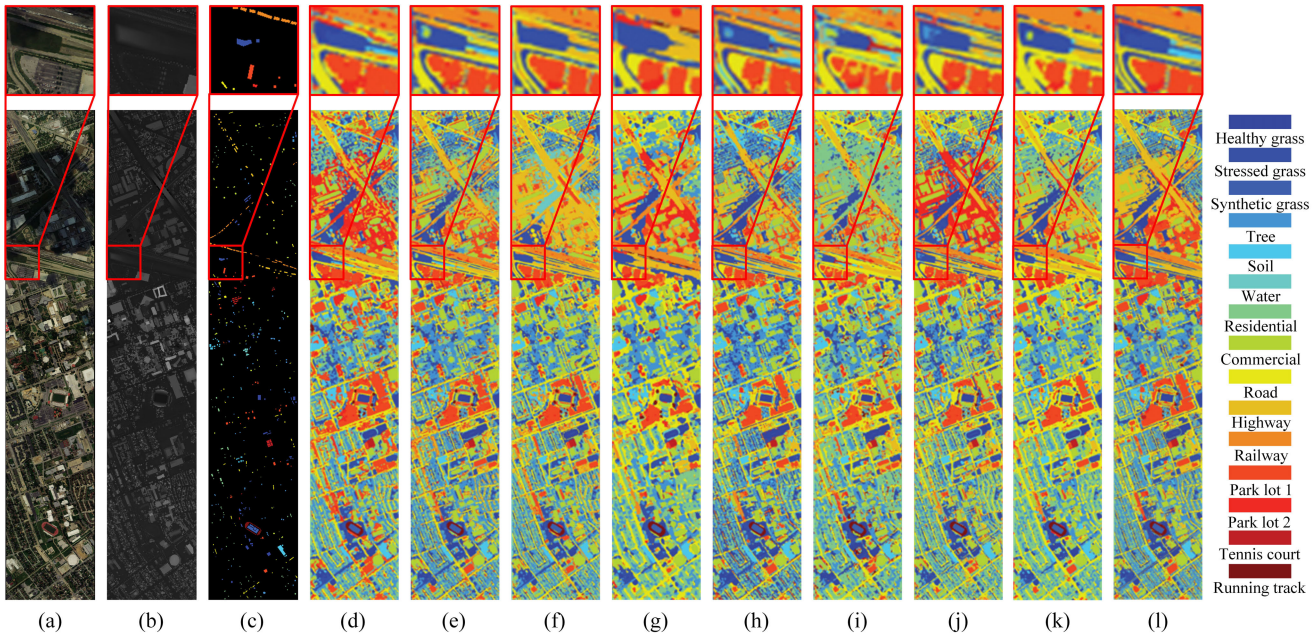


Fig. 3. Comparative classification maps for the Houston2013 dataset. (a) HSI false-color representation (R: band 64, G: band 43, and B: band 20). (b) LiDAR-based digital surface model (DSM). (c) Ground-truth map. (d) CoupledCNN (94.36%). (e) GLT-Net (95.09%). (f) AMSSE-Net (90.90%). (g) ExViT (93.38%). (h) HFSNet (91.92%). (i) M2FNet (93.78%). (j) MSHT (92.47%). (k) GFSFN (95.38%). (l) PS2F2L (95.70%).

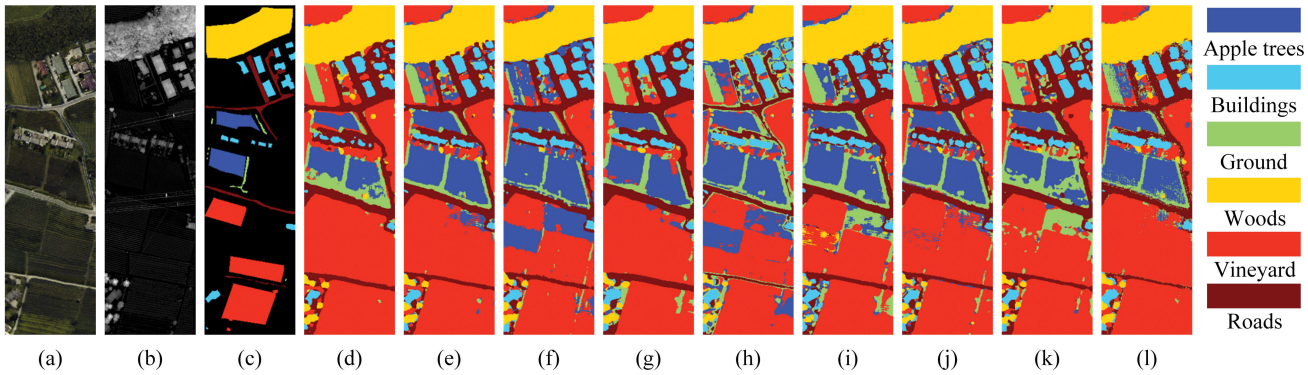


Fig. 4. Comparative classification maps for the Trento dataset. (a) HSI false-color representation (R: band 20, G: band 15, and B: band 5). (b) LiDAR-based DSM. (c) Ground-truth map. (d) CoupledCNN (99.13%). (e) GLT-Net (99.49%). (f) AMSSE-Net (98.82%). (g) ExViT (99.00%). (h) HFSNet (98.78%). (i) M2FNet (99.17%). (j) MSHT (99.14%). (k) GFSFN (99.46%). (l) PS2F2L (99.54%).

TABLE II
ABLATION STUDY OF THE PROPOSED MODEL WITH AND WITHOUT S2FL AND SF2L

Methods	Houston2013			Trento			MUUFL		
	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
w/o spatial-frequency	93.66	94.57	93.14	97.11	97.25	96.17	86.45	88.68	82.60
w/o spatial-spectral	94.20	95.00	93.73	97.59	97.43	96.80	86.07	88.18	82.10
PS2F2L	**95.70**	**96.34**	**95.36**	**99.54**	**99.01**	**99.39**	**89.10**	**89.50**	**85.82**

categories. More detailed information on these three datasets can be found in [37], [38], and [39].

The model ablation results for the three datasets are shown in Table II, and the parameter experiments are illustrated in Fig. 2.

2) Experimental Setting: In our experiments, the proposed PS2F2L method was implemented in the PyTorch framework. The experiments were performed on a computer with an Intel Core i7-10700 processor (2.90 GHz) and an NVIDIA GeForce RTX 2060 GPU. We used the Adam optimizer with a batch

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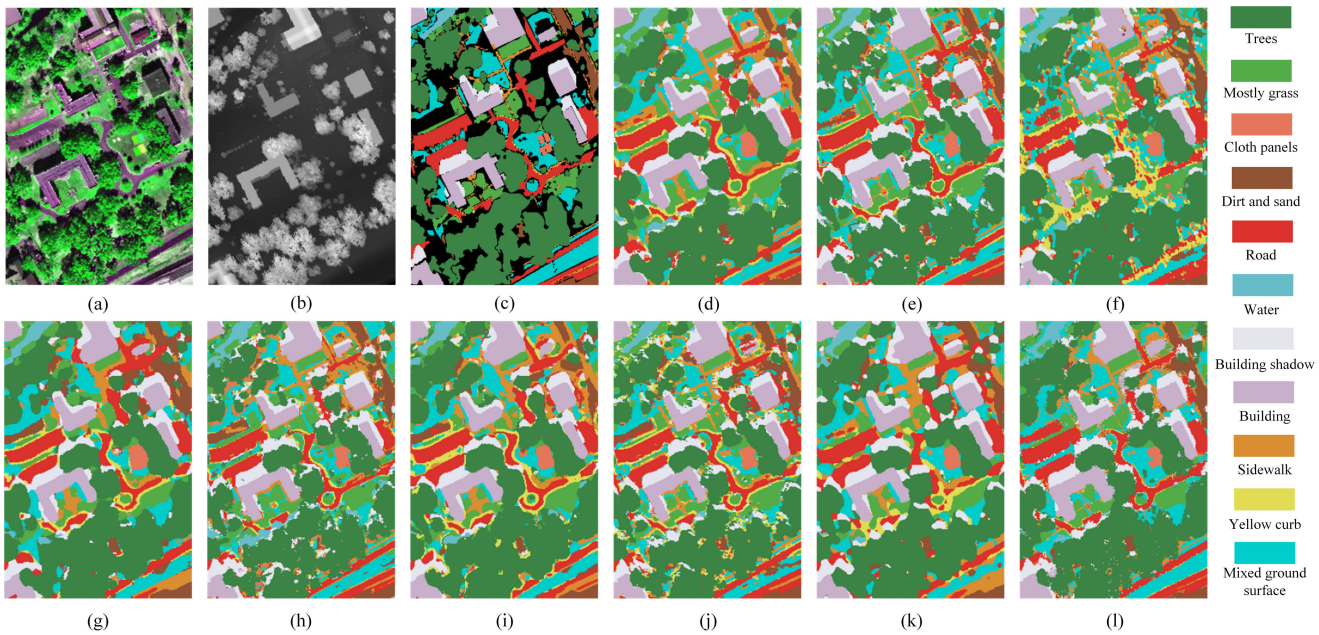


Fig. 5. Comparative classification maps for the MUUFL dataset. (a) HSI false-color representation (R: band 30, G: band 50, and B: band 21). (b) LiDAR-based DSM. (c) Ground-truth map. (d) CoupledCNN (84.39%). (e) GLT-Net (84.80%). (f) AMSSE-Net (75.24%). (g) ExViT (80.18%). (h) HFSNet (83.18%). (i) M2FNet (82.80%). (j) MSHT (83.38%). (k) GFSFN (83.98%). (l) PS2F2L (89.10%).

TABLE III

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

No.	Class (Train/Test)	Performance								
		CoupledCNN	GLT-Net	AMSSE-Net	ExViT	HFSNet	M2FNet	MSHT	GFSFN	PS2F2L
1	Healthy grass (20/1231)	94.02±4.51	92.71±3.70	93.43±4.84	92.78±3.61	93.50±4.05	92.69±3.77	93.98±0.09	93.70±4.85	**94.82±4.45**
2	Stressed grass (20/1234)	96.84±3.01	96.52±3.25	96.63±1.46	96.15±3.03	95.28±2.76	96.80±3.60	89.95±0.08	96.18±2.67	**96.98±2.85**
3	Synthetic grass (20/677)	99.46±0.84	99.73±0.32	99.51±0.70	99.00±1.33	99.85±0.14	99.73±0.17	99.43±0.05	99.76±0.38	**100.00±0.00**
4	Tree (20/1224)	98.33±1.68	97.84±2.47	98.75±1.76	96.06±2.43	97.09±2.60	95.89±2.30	**99.03±0.03**	98.06±2.45	97.76±2.56
5	Soil (20/1222)	99.63±0.41	99.81±0.27	99.47±0.60	99.12±1.69	99.43±0.46	97.44±1.38	**100.00±0.00**	99.92±0.11	99.93±0.08
6	Water (20/305)	97.31±3.82	96.78±4.11	92.26±4.84	95.87±4.54	92.32±5.13	96.72±2.93	**99.40±0.13**	98.00±3.85	98.26±3.80
7	Residential (20/1248)	94.81±2.59	95.36±1.85	88.11±6.11	92.70±3.82	92.78±3.31	94.02±1.97	94.61±0.05	**96.85±2.47**	95.68±2.24
8	Commercial (20/1224)	87.05±5.54	**91.30±2.80**	81.88±5.03	86.09±7.51	85.84±6.97	89.40±4.74	89.03±0.10	88.52±9.12	89.70±3.37
9	Road (20/1232)	86.74±4.04	94.18±1.43	74.32±9.34	91.09±2.85	80.87±5.18	88.68±3.57	88.35±0.11	**94.29±1.87**	91.11±3.39
10	Highway (20/1207)	93.61±3.20	91.48±3.17	89.80±4.71	89.25±4.48	83.14±3.23	92.17±1.73	83.22±0.13	90.48±4.86	**95.47±2.62**
11	Railway (20/1215)	95.57±1.77	92.88±4.05	90.67±4.98	93.50±3.72	91.90±7.89	93.95±3.48	88.75±0.08	96.47±1.60	**96.99±0.78**
12	Parking lot 1 (20/1213)	87.44±4.75	90.79±3.23	84.78±5.07	86.42±4.94	88.73±3.74	87.26±4.73	83.68±0.21	91.19±4.43	**91.48±3.81**
13	Parking lot 2 (20/449)	98.04±1.66	97.57±2.04	87.79±5.23	96.93±2.81	90.13±4.22	96.45±2.18	97.41±0.10	**98.17±1.97**	97.03±1.70
14	Tennis court (20/408)	99.97±0.07	99.77±0.51	99.50±0.84	99.66±0.57	99.85±0.22	99.43±0.82	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	Running track (20/640)	99.67±0.73	99.95±0.10	99.31±0.61	99.97±0.09	99.56±0.49	99.70±0.52	**100.00±0.00**	99.96±0.09	99.90±0.18
	OA (%)	94.36±0.94	95.09±0.59	90.90±1.64	93.38±0.98	91.92±1.55	93.78±0.63	92.47±0.04	95.38±1.11	**95.70±0.79**
	AA (%)	95.23±0.88	95.78±0.48	91.75±1.44	94.30±0.88	92.68±1.24	94.65±0.50	93.79±0.04	96.11±0.89	**96.34±0.68**
	Kappa (%)	93.90±1.01	94.70±0.64	90.16±1.77	92.84±1.06	91.26±1.68	93.28±0.68	91.86±0.04	95.01±1.20	**95.36±0.85**

TABLE IV

CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE TRENTO DATASET

No.	Class (Train/Test)	Performance								
		CoupledCNN	GLT-Net	AMSSE-Net	ExViT	HFSNet	M2FNet	MSHT	GFSFN	PS2F2L
1	Apple trees (20/4014)	98.72±0.44	99.83±0.15	98.85±0.99	98.51±0.94	98.56±0.90	99.53±0.37	99.65±0.28	99.38±0.30	**99.94±0.06**
2	Buildings (20/2883)	97.20±1.03	97.93±0.66	95.29±4.59	96.34±2.72	97.96±1.06	96.60±1.90	98.52±0.69	98.51±0.67	**98.75±0.67**
3	Ground (20/459)	98.12±1.63	98.08±1.23	95.55±4.21	**98.76±1.51**	96.36±2.32	98.10±1.30	99.12±0.94	98.25±1.20	98.16±1.04
4	Woods (20/9103)	99.91±0.10	**100.00±0.00**	99.99±0.01	**100.00±0.00**	99.94±0.09	99.91±0.15	99.99±0.32	**100.00±0.00**	**100.00±0.00**
5	Vineyard (20/10481)	**99.99±0.01**	99.98±0.01	99.38±1.44	99.76±0.36	99.19±0.55	99.97±0.04	99.78±0.18	99.94±0.07	99.97±0.04
6	Roads (20/3154)	96.46±1.15	**97.57±1.30**	97.30±1.39	96.69±2.62	95.50±1.82	96.45±2.37	94.45±0.30	97.40±1.96	97.21±1.24
	OA (%)	99.13±0.12	99.49±0.11	98.82±0.61	99.00±0.13	98.78±0.28	99.17±0.20	99.14±0.32	99.46±0.22	**99.54±0.09**
	AA (%)	98.40±0.25	98.90±0.31	97.73±0.98	98.34±0.28	97.92±0.52	98.43±0.46	98.59±0.50	98.92±0.41	**99.01±0.19**
	Kappa (%)	98.84±0.17	99.32±0.15	98.43±0.81	98.66±0.18	98.38±0.37	98.90±0.27	98.85±0.43	99.28±0.30	**99.39±0.13**

size of 64 and an initial learning rate of 0.001, and trained the model for 200 epochs.

3) Evaluation Metrics: We assessed model performance through multiple evaluation metrics, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient. To ensure robust performance evaluation unaffected by sampling bias, all quantitative results were reported as averages across ten independent runs.

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TABLE V
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE MUUFL DATASET

No.	Class (Train/Test)	Performance								
		CoupledCNN	GLT-Net	AMSSE-Net	ExViT	HFSNet	M2FNet	MSHT	GFSFN	PS2F2L
1	Trees (20/23226)	90.04±1.49	88.57±3.32	81.08±5.05	84.85±2.85	86.66±2.79	90.64±1.66	86.73±0.00	88.54±1.52	**92.11±1.64**
2	Mostly grass (20/4250)	74.78±5.12	77.82±7.05	70.13±8.70	73.35±3.91	75.97±9.87	68.70±4.28	76.02±0.04	76.48±4.28	**84.40±4.74**
3	Mixed ground surface (20/6862)	66.13±4.04	66.87±8.18	52.31±9.39	60.32±0.65	66.05±6.04	62.31±6.15	62.01±0.04	68.06±4.28	**75.82±6.32**
4	Dirt and sand (20/1806)	87.87±6.30	85.04±6.06	76.49±8.23	87.44±3.99	84.25±6.51	80.86±6.63	91.81±0.05	90.83±4.37	**92.77±4.78**
5	Road (20/6667)	81.53±4.96	86.16±3.65	67.81±6.01	74.63±4.55	85.16±4.92	80.25±4.83	86.59±0.02	78.66±5.50	**90.31±2.03**
6	Water (20/446)	99.43±0.57	98.78±1.47	98.14±2.14	99.15±0.86	99.32±0.80	99.08±1.30	**100.00±0.00**	99.79±0.33	99.59±0.35
7	Building shadow (20/2213)	86.69±3.39	88.72±4.60	80.60±6.55	87.26±3.71	86.83±4.31	85.82±3.71	89.17±0.06	**90.48±3.52**	89.79±3.45
8	Building (20/6220)	**95.32±1.54**	94.89±1.05	90.70±3.11	93.36±1.76	93.88±1.61	92.81±3.16	93.90±0.01	94.84±1.23	95.27±1.78
9	Sidewalk (20/1365)	59.36±6.85	67.04±5.68	52.18±10.49	59.49±4.56	60.37±6.22	56.00±5.37	65.87±0.20	59.38±5.93	**73.75±4.08**
10	Yellow curb (20/163)	80.73±5.87	82.63±5.31	69.57±6.87	76.75±6.41	74.96±11.49	59.44±9.77	76.01±1.51	78.15±7.10	**94.41±3.13**
11	Cloth panels (20/249)	95.78±2.15	95.66±2.03	94.49±4.32	95.26±2.31	95.30±2.83	95.62±2.01	94.29±0.39	**97.83±1.94**	96.22±1.63
	OA (%)	84.39±0.84	84.80±1.34	75.24±2.64	80.18±1.94	83.18±1.30	82.80±0.64	83.38±0.02	83.98±1.08	**89.10±0.83**
	AA (%)	83.42±0.97	84.74±0.95	75.77±2.23	81.07±1.03	82.61±1.18	79.23±1.03	83.85±0.17	83.92±1.05	**89.50±0.64**
	Kappa (%)	79.79±1.04	80.42±1.58	68.72±3.08	74.75±2.31	78.39±1.56	77.71±0.76	78.73±0.02	79.37±1.32	**85.82±1.05**

B. Parameter Selection

Different patch sizes imply varying spatial scales of land cover information, while different wavelet basis functions also affect the effectiveness of SF2L. Based on the above considerations, we designed separate experiments to investigate the effects of varying patch sizes and wavelet basis functions. Fig. 2 shows the results of parameter experiments on the Houston2013, Trento, and MUUFL datasets, where the color scale represents different OA ranges, with red and blue indicating the highest and lowest OA values, respectively. As shown in Fig. 2, the proposed model exhibits certain sensitivity to different patch sizes and wavelet basis functions on the same dataset. This indicates that selecting appropriate hyperparameters is crucial for improving the model performance.

1) Patch Size: In this experiments, we tested five patch sizes as $\mathbf{p} = \{6, 8, 10, 12, 14\}$ while maintaining identical settings for all other parameters. It is obvious from Fig. 2 that the OA values of the three datasets reach the highest values at $\mathbf{p} = 12$. When a small patch size ($\mathbf{p}$) is selected, the available receptive field shrinks, which limits the model's access to contextual information. Conversely, an excessively large patch size tends to introduce redundant information, leading to the model's difficulty in capturing discriminable local features. Both scenarios ultimately affect the overall classification accuracy. Therefore, we set the patch size of the three datasets to 12.

2) Wavelet Basis Functions: To assess the impact of wavelet selection, we conducted experiments using four basis functions including haar, db2, sym2, and coif2. Compared to haar, sym2, and coif2, the db2 exhibits optimal classification performance on all three datasets (see Fig. 2). This is mainly attributed to the fact that the haar wavelet, which has only the first-order vanishing moments, can capture signal mutation features such as edges, but has relatively poor representation of smooth regions. Although the sym2 wavelet improves edge localization accuracy by optimizing symmetry, it may trade off some frequency selectivity to achieve higher symmetry. Coiflet wavelets, despite possessing larger vanishing moments, may suffer from extended support lengths that lead to overfitting or amplified boundary effects in finite samples, thereby weakening their ability to extract discriminative features. With its second-order vanishing moments, the db2 wavelet can better capture smooth regions, suppress high-frequency noise, and extract more accurate feature details, while maintaining excellent frequency component discrimination. Therefore, the wavelet basis function was set as db2 for all three datasets.

C. Ablation Studies

To verify the positive impact of spatial–spectral and spatial–frequency features fusion on the HSI–LiDAR joint classification task, we designed ablation experiments including three test cases. Table II provides corresponding results based on three evaluation metrics (i.e., OA, AA, and Kappa) for the three datasets. In the case of w/o (i.e., without) spatial–frequency, only spatial–spectral features $\mathbf{F_{ss}}$ were fed into the classifier. In the case of w/o spatial–spectral, spatial–frequency features $\mathbf{F_{sf}}$ were solely used for classification. In PS2F2L, both spatial–spectral information and spatial–frequency information were integrated through spatial–spectral–frequency interactive fusion and DLF. As can be seen in Table II, all three metrics improve across all datasets after fusing spatial–spectral and spatial–frequency information. More specifically, the integration of spatial–spectral–frequency features increases OA by 2.04% and 1.50% for the Houston2013 dataset, 2.43% and 1.95% for the Trento dataset, and 2.65% and 3.03% for the MUUFL dataset. These results not only validate the contributions of different domains to classification performance but also prove that the proposed PS2F2L maximizes their complementary potential for superior classification.

D. Performance Comparison

To further demonstrate the effectiveness of the proposed PS2F2L model, we conducted comprehensive comparative experiments with the SOTA methods across three dimensions: classification performance, stability under varying training sample volumes, and computational complexity. The selected SOTA approaches included CoupledCNN [14], GLT-Net [40], AMSSE-Net [41], ExViT [33], HFSNet [42], M2FNet [20], MHST [43], and GFSFN [29]. Among them, CoupledCNN, AMSSE-Net, and MHST employ CNN, while GLT-Net, ExViT, HFSNet, and M2FNet utilize a transformer for feature extraction and fusion in the image domain. Whereas GFSFN operates as a spatial–frequency-domain fusion approach. All methods were implemented on the same platform using the

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same training and testing samples. The hyperparameters for all compared methods were set according to the recommendations in their corresponding articles, enabling them to achieve their best comprehensive performance in our experiments.

1) Classification Performance: Both quantitative and qualitative evaluations were used for HSI–LiDAR joint classification comparisons. To ensure statistical robustness, all experimental results were derived from ten independent trial runs, with each metric reported as the mean ± standard deviation. The quantitative results on three test datasets are presented in Tables III–V, with the best results shown in bold. The corresponding visual comparisons are shown in Figs. 3–5, including HSI false-color images, LiDAR data visualization, ground-truth label, and classification maps from different comparison methods.

For the Houston2013 dataset, the proposed PS2F2L achieves optimal performance on OA, AA, and Kappa, three evaluation metrics, and attains the highest accuracy in nearly half (7/15) of the object classes, as shown in Table III. Compared to the second-best method (GFSFN), PS2F2L improves OA, AA, and Kappa by 0.32%, 0.23%, and 0.35%, respectively. The correct rates of classifying synthetic grass and tennis court classes are 100%. According to the visual comparison results from Fig. 3, the proposed method maintains superior boundary delineation and linear continuity in transportation infrastructure classification (roads, highways, and railroads) compared to the other SOTA methods. Even partially obscured by cloud shadows, the classified highway shows no breaks or discontinuities in its path, which demonstrates the stability and robustness of our PS2F2L model. Additionally, in locally magnified regions, methods that rely solely on the spatial domain generally suffer from unnatural classification boundaries or category misclassification. Although the GFSFN method, which combines the spatial and frequency domains, performs better in boundary delineation, it still cannot avoid misclassification problems. In contrast, our method can more accurately restore the complete contour of the synthetic grass and clearly depict the boundary details. This is mainly due to the gradual fusion strategy and multidomain joint approach, which enable PS2F2L to more comprehensively integrate multimodal and multidomain discriminative information, thereby improving classification capabilities.

On the Trento dataset, the proposed PS2F2L also outperforms all the comparison methods in OA, AA, and Kappa index evaluations. With only six categories and relatively low scene complexity, PS2F2L approaches the theoretical accuracy ceiling on the Trento dataset, attaining 99.54% average OA and top accuracy in half of the classes (see Table IV). This near-perfect performance stems from the dataset's elementary structure featuring minimal class overlap and simple spatial distributions. Despite the predominance of regularly distributed ground objects in this scene, PS2F2L displays better discriminative capability for mixed surfaces particularly at boundary transition zones, as shown in Fig. 4.

Similarly, the proposed method maintains superior performance on the MUUFL dataset. As displayed in Table V, PS2F2L surpasses all competitors by a significant margin in OA, AA, and Kappa scores. Relative to the runner-up GLT-Net, it shows 4.30%, 4.76%, and 5.40% average advantages in these three measures, respectively. Meanwhile, the proposed method secures top rankings in the majority (7/11) of the object classes, which demonstrates its excellent discriminative capability. Comparative visualization in Fig. 5 also confirms PS2F2L's enhanced discriminative capability through more accurate boundary delineation in complex heterogeneous regions (see roads and sidewalks under the trees). This is because our PS2F2L model introduces the multidomain joint feature learning strategy and the progressive fusion framework, enabling it to more effectively capture the structural features of irregular objects and construct more representative feature representations compared to traditional spatial-domain methods.

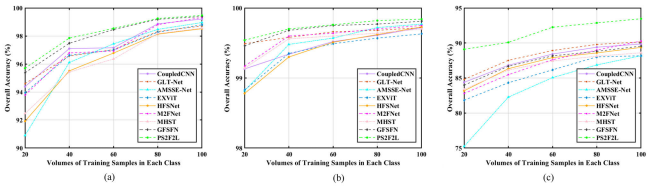


Fig. 6. Classification performance comparison under different training sample sizes on (a) Houston2013, (b) Trento, and (c) MUUFL.

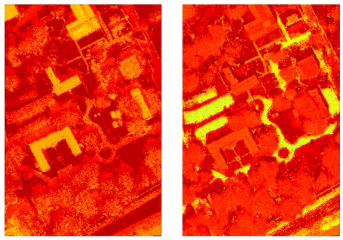
2) Stability Under Varying Training Sample Volumes: To further validate the stability and robustness of the proposed PS2F2L, we conducted multiple classification experiments with training sample sizes ranging from 20 to 100 samples per class (in increments of 20 samples). For each sample size configuration, a fixed number of samples were randomly selected for training, with the remaining samples used for testing. All reported results represent the average of ten independent trials with identical training/testing sample sets to ensure statistical reliability. The classification performance of the proposed PS2F2L model and other comparison methods across varying training sample sizes is presented in Fig. 6, using OA as the evaluation metric. Obviously, all methods exhibit monotonically increasing classification accuracy with more training samples. This indicates that an increase in training samples helps improve the classification performance of DL classification models. More importantly, PS2F2L consistently performs the best across all training set sizes, with its advantage being particularly pronounced when training samples are scarce (e.g., 20 per class). This is because DWT helps to capture crucial high-frequency features (related to edges and corners) in the network learning, which are particularly important for classification tasks. In addition, the multistage fusion mechanism that effectively integrates spatial–spectral and spatial–frequency features from HSI and LiDAR data is also a key reason for the improved performance.

3) Function of Subtraction-Based Attention Mechanism: To validate the effectiveness and advantages of the proposed subtraction-based attention mechanism in multimodal feature fusion for HSI–LiDAR images, we conducted a visual analysis of the generated attention maps (see Fig. 7). As shown, the two attention maps output by this mechanism precisely focus on key discriminative regions within the spatial–spectral feature ($\mathbf{F}_{\mathbf{ss}}$) and spatial–frequency feature ($\mathbf{F}_{\mathbf{sf}}$), respectively. This fully demonstrates that the designed attention mechanism,

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TABLE VI
MODEL COMPLEXITY AND TRAINING TIME (S) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Index	CoupledCNN	GLT-Net	AMSSE-Net	ExViT	HFSNet	M2FNet	MSHT	GFSFN	PS2F2L
Trainable Params (K)	101.29	599.97	1057.55	239.91	419.76	282.09	1164.458	1354.44	166.814
Computation Cost (MFLOPs)	2.29	39.07	28.89	45.84	14.45	90.64	79.77	84.23	7.90
Training time (s)	107.10	1048.18	83.30	1087.63	1940.68	8692.09	3663.00	121.91	123.34



(a) (b)

Fig. 7. Visualization of normalized attention maps: (a) $\mathbf{w_1}$ and (b) $\mathbf{w_2}$.

by explicitly modeling cross-domain differences between spatial–spectral and spatial–frequency features, can effectively guide the network to focus on important yet subtle information. Consequently, it facilitates adaptive fusion while directionally enhancing the discriminative capacity of the fused features.

4) Computational Complexity Analysis: By using the Houston2013 dataset as an example, the model complexity of different networks was measured, including the number of trainable parameters updated during backpropagation and the computational cost measured in millions of floating point operations per second (MFLOPs). To conduct a more comprehensive analysis, the training times of different methods were also recorded. As shown in Table VI, PS2F2L exhibits the second-lowest parameter count and MFLOPs among all comparison models (after CoupledCNN). Although the proposed model does not achieve the lowest computational complexity, it maintains competitive efficiency rankings while delivering the best classification performance (as evidenced in Tables III–V). This demonstrates that the proposed method successfully achieves an optimal performance–efficiency balance through its multistage progressive fusion strategy, ensuring both technical feasibility and practical value for real-world deployment.

V. CONCLUSION

This article presents a PS2F2L network, a lightweight yet effective framework for joint HSI and LiDAR classification. The proposed method systematically addresses modal discrepancies through a novel three-stage progressive fusion pipeline: 1) multimodal feature extraction; 2) cross-domain interactive enhancement; and 3) adaptive classification refinement. The framework first employs a dual-branch architecture to simultaneously capture complementary features—one branch obtains spectral–spatial information through 1-D/2-D convolutions while the other extracts spatial–frequency characteristics via wavelet decomposition, effectively overcoming single-domain perception limitations. An innovative interactive fusion module then bridges these domains by strategically leveraging their representational differences to enhance feature discriminability. The pipeline concludes with adaptive DLF to optimize classification stability. Designed with computational efficiency in mind, PS2F2L achieves comprehensive multimodal integration through its streamlined workflow. Extensive evaluations on three benchmark datasets demonstrate consistent superiority over SOTA methods in both quantitative metrics and qualitative assessments. Future work will focus on developing sample selection strategies with uncertainty awareness to enhance annotation efficiency, as well as exploring cross-scene hyperspectral classification methods based on domain adaptation or unsupervised learning.

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